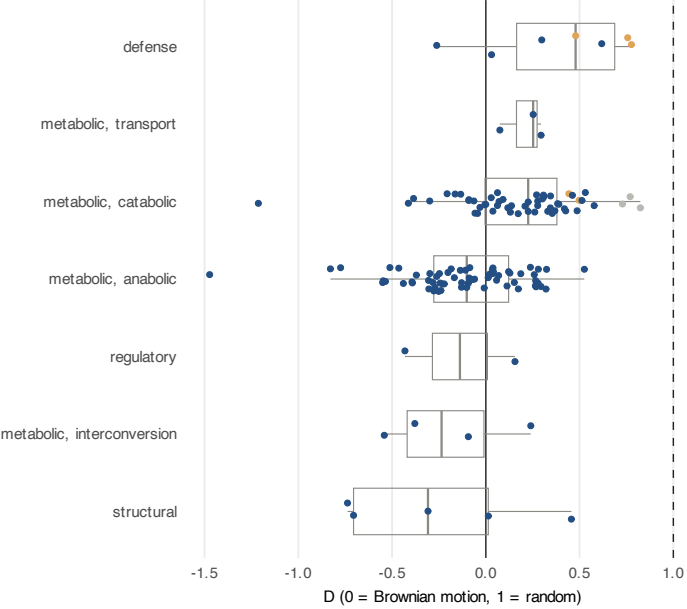


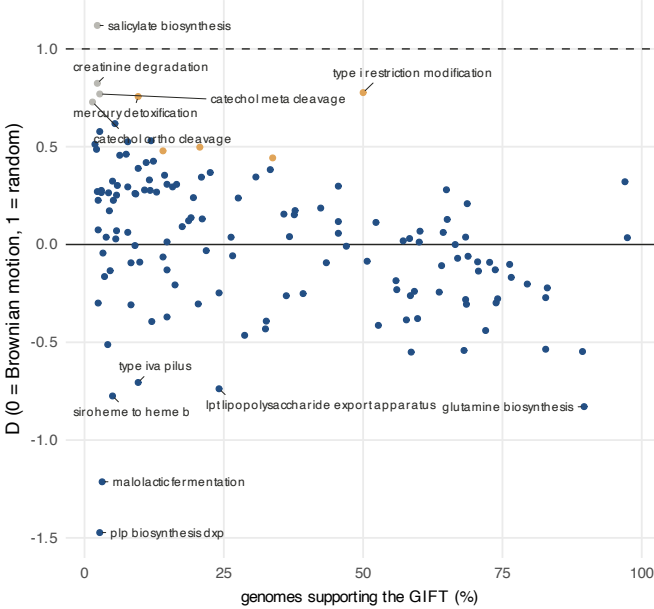
How each GIFT is distributed over the phylogeny

Fritz and Purvis's D for the 140 GIFTs carried and lacked by at least 10 of 696 genomes, on the pruned GTDB bac120 tree (1,000 permutations; categories at a false discovery rate of 0.05). Lower D means the GIFT is confined to related genomes.

a By type and metabolic mode



b Against prevalence



• as conserved as Brownian motion • between Brownian motion and random • not distinguishable from random

D describes how an encoded, curated capability is distributed over this enriched panel; it is not an ancestral-state reconstruction and does not measure gain or loss rates.